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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Time: 03 Hours

Code: 100606 ***Subject: Digital Signal Processing*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[14]**

(a) Which of the following is a continuous-time signal?
(i) Sinusoidal wave
(ii) Impulse signal
(iii) Unit step signal
(iv) All of the above

(b) What does DTFT stand for?
(i) Discrete-Time Fourier Transform
(ii) Digital-Time Fourier Transform
(iii) Dynamic-Time Fourier Transform
(iv) Discrete Transfer Fourier Transform

(c) The process of converting continuous-time signals into discrete-time signals is called?
(i) Quantization
(ii) Sampling
(iii) Lasing
(iv) Reconstruction

(d) What is the frequency domain representation of LTI systems called?
(i) Fourier Series
(ii) Fourier Transform
(iii) Laplace Transform
(iv) Both i and ii

(e) A system is causal if:
(i) Output depends on future inputs
(ii) Output depends only on present and past inputs
(iii) Output is independent of input
(iv) None of the above

(f) What is the result of convolving two impulse signals?
(i) Impulse signal
(ii) Zero signal
(iii) Step signal
(iv) Constant signal

(g) The sampling theorem states that to avoid aliasing, the sampling rate must be:
(i) Less than the Nyquist rate
(ii) Equal to the Nyquist rate
(iii) Greater than or equal to twice the maximum signal frequency
(iv) Greater than the Nyquist rate

(h) Which of the following is used to compute the convolution of two discrete signals?
(i) Fourier Transform
(ii) Laplace Transform
(iii) Z-transform
(iv) Convolution sum

(i) Which of the following represents the Nyquist rate?
(i) Twice the highest frequency
(ii) Half the highest frequency
(iii) Equal to the highest frequency
(iv) None of the above

(j) Which property of DFT leads to a reduction in computational complexity?
(i) Symmetry
(ii) Linearity
(iii) Circular convolution
(iv) Time-shifting

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Q.2 Solve both questions : **[14]**

(a) Explain the advantages of DSP over analog signal processing. Provide practical examples to support your explanation.

(b) Discuss the stability and causality conditions of discrete-time systems. Provide examples of systems that are stable and causal.

Q.3 Solve both questions : **[14]**

(a) Explain the basic elements of a DSP system with a block diagram. How does it function in practical applications?

(b) Compare the correlation and convolution of discrete-time signals with examples.

Q.4 Solve both questions : **[14]**

(a) Differentiate between continuous-time and discrete-time signals. Use diagrams to illustrate their key characteristics.

(b) Explain the properties of the Discrete-Time Fourier Transform (DTFT) for LTI system with applications.

Q.5 Solve both questions : **[14]**

(a) Derive the formula for the Inverse Discrete-Time Fourier Transform with suitable examples.

(b) Explain the process of sampling continuous-time signals and discuss the significance of the Nyquist rate in digital signal processing.

Q.6 Solve both questions : **[14]**

(a) Discuss the relationship between the time-domain and frequency-domain representations of signals using the Fourier transform.

(b) Explain the reconstruction of signals from their samples using an ideal low-pass filter.

Q.7 Solve both questions : **[14]**

(a) Explain the difference between FIR and IIR filters. What are the advantages and disadvantages of each?

(b) Discuss the properties of the Discrete Fourier Transform (DFT). Explain its significance.

Q.8 Solve both questions : **[14]**

(a) Derive the Fast Fourier Transform (FFT) algorithm and explain how it improves the computation of the DFT.

(b) Determine the convolution for the two sequences $x(n)=\{3,2,1,2\}$, $h(n)=\{1,2,1,2\}$.

Q.9 Solve both questions : **[14]**

(a) Illustrate how the properties of linearity and time-invariance simplify the analysis of discrete-time systems with examples.

(b) Explain the concept of periodicity in continuous-time and discrete-time signals. How do their periodicity conditions differ?

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